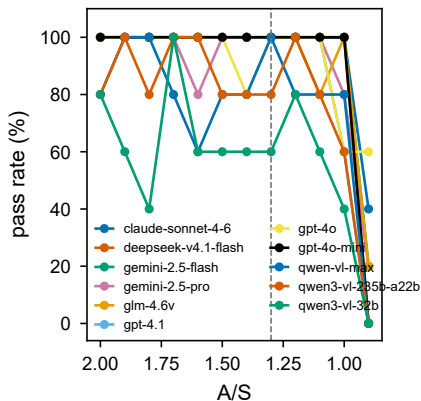
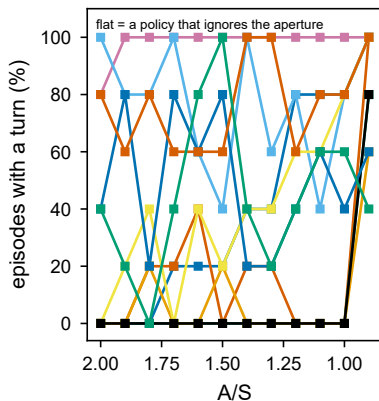


The eleven models together

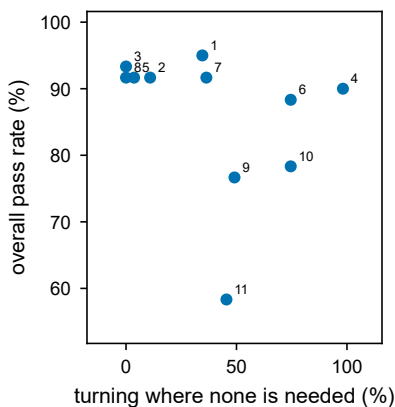
pass rate per aperture



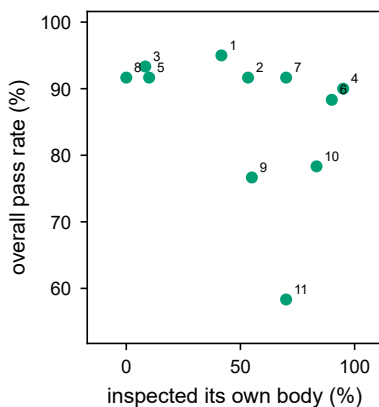
turn rate per aperture



spurious rotation vs success



self-inspection vs success



Points in both scatters, numbered: 1 claude-sonnet-4-6 (turn on demand) 2 deepseek-v4.1-flash (turn on demand) 3 gemini-2.5-flash (straight walker) 4 gemini-2.5-pro (fixed-policy turner) 5 glm-4.6v (straight walker) 6 gpt-4.1 (fixed-policy turner) 7 gpt-4o (turn on demand) 8 gpt-4o-mini (straight walker) 9 qwen-vl-max (mixed) 10 qwen3-vl-235b-a22b (fixed-policy turner) 11 qwen3-vl-32b (mixed)

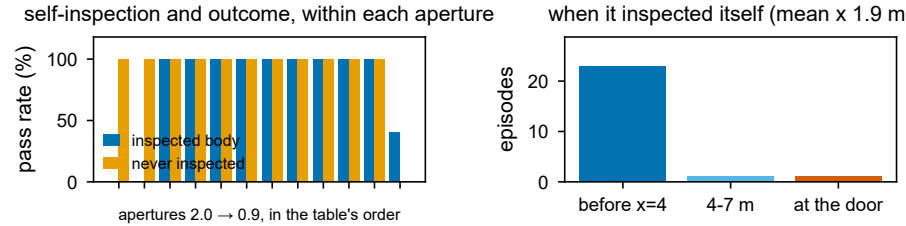
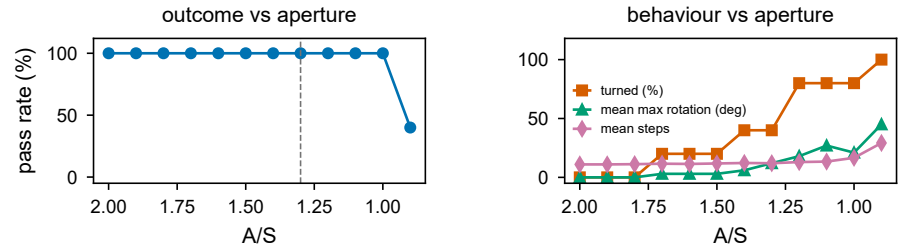
Every model, sorted by overall pass rate
pass rate at the two apertures that need rotation (A/S 1.0, 0.9); wide turn% = turning where none

model	type	pass%	narrow	turn%	wide turn%	first turn x	look%	pass look	pass nolook
claude-sonnet-4-	turn on demand	95	7/10	40	35	2.31	42	88	100
gemini-2.5-flash	straight walker	93	6/10	7	0	7.06	8	60	96
deepseek-v4.1-fl	turn on demand	92	5/10	18	11	3.77	53	84	100
glm-4.6v	straight walker	92	6/10	8	4	5.00	10	50	96
gpt-4o	turn on demand	92	6/10	42	36	1.64	70	88	100
gpt-4o-mini	straight walker	92	5/10	7	0	7.25	0	–	92
gemini-2.5-pro	fixed-policy turner	90	5/10	98	98	1.99	95	89	100
gpt-4.1	fixed-policy turner	88	3/10	77	75	1.71	90	87	100
qwen3-vl-235b-a2	fixed-policy turner	78	3/10	77	75	1.82	83	76	90
qwen-vl-max	mixed	77	4/10	50	49	2.52	55	58	100
qwen3-vl-32b	mixed	58	2/10	45	45	1.22	70	48	83

Pass rate spans 58-95%, and the two apertures that need rotation already separate the models (claude-sonnet- 7/10, gemini-2.5-fla 6/10, deepseek-v4.1- 5/10).

Turning where no turn is needed averages 39% and ranges from 0% (gemini-2.5-fla) to 98% (gemini-2.5-pro), so the models differ by more than an order of magnitude on a decision the aperture does not require at all. Inspecting its own body, compared inside each aperture rather than pooled (pooling would be confounded, because looking down clusters at the narrow Levels): -9.1 percentage points on average, positive in 1 of 11 models, so inspection does not buy success by itself.

The reasoning names the opening, the turn and the model's own width; what fails is the amount of rotation and the alignment, not knowing that a turn is needed.



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	11.0	0	0.0	0	—	100
1.9	1.083	5/5	0	0	11.0	0	0.0	0	—	100
1.8	1.026	5/5	0	0	11.2	0	0.2	20	100	100
1.7	0.969	5/5	20	0	11.6	3	0.6	20	100	100
1.6	0.912	5/5	20	0	11.4	3	0.4	20	100	100
1.5	0.855	5/5	20	0	11.8	3	0.8	60	100	100
1.4	0.798	5/5	40	0	12.2	6	1.2	60	100	100
1.3	0.741	5/5	40	20	12.0	12	1.0	20	100	100
1.2	0.684	5/5	80	0	13.0	18	2.0	60	100	100
1.1	0.627	5/5	80	40	13.4	27	2.4	60	100	100
1.0	0.570	5/5	80	0	16.6	21	5.6	80	100	100
0.9	0.513	2/5	100	100	29.2	45	14.2	100	40	—

Type: turn on demand. whether it turns follows the aperture (only 35% of the no-turn episodes turn, first turn at x = 2.31 m on average), which is the closest any of these models comes to a judgement rather than a fixed action; it passes 7/10 of the two rotation levels.

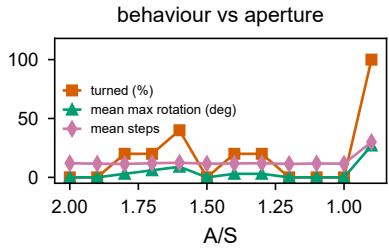
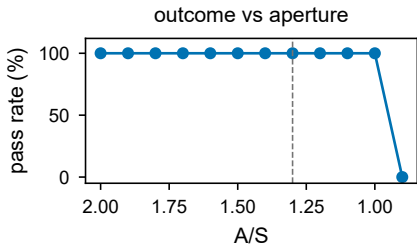
Reasoning mentions (share of episodes): opening 100%, align 100%, turn 92%, width 70%, sideways 40%, blocked 28%, budget 3%

Actions: forward 84.9%, turn_left 5.2%, look_down 3.8%, turn_right 2.7%, right 1.8%, left 1.3%

Strategy labels: FRONTAL 60%, ROT_EARLY 35%, ROT_MID 3%, ROT_LATE 2%

It turned in 40% of all episodes (first turn at x = 2.3 m on average), and in 35% of the episodes at apertures that need no turn at all.

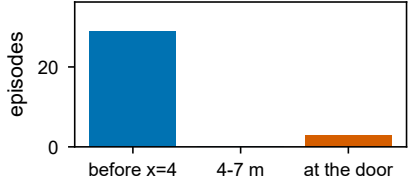
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 9 apertures positive).



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.6 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	12.0	0	1.0	80	100	100
1.9	1.083	5/5	0	0	11.6	0	0.6	60	100	100
1.8	1.026	5/5	20	0	11.4	3	0.4	20	100	100
1.7	0.969	5/5	20	0	12.0	6	1.0	60	100	100
1.6	0.912	5/5	40	0	12.4	9	1.4	60	100	100
1.5	0.855	5/5	0	0	11.6	0	0.6	60	100	100
1.4	0.798	5/5	20	0	11.8	3	0.8	40	100	100
1.3	0.741	5/5	20	0	12.0	3	1.0	20	100	100
1.2	0.684	5/5	0	0	11.4	0	0.4	40	100	100
1.1	0.627	5/5	0	0	11.8	0	0.8	60	100	100
1.0	0.570	5/5	0	0	11.6	0	0.6	40	100	100
0.9	0.513	0/5	100	0	30.0	27	15.0	100	0	-

Type: turn on demand. whether it turns follows the aperture (only 11% of the no-turn episodes turn, first turn at x = 3.77 m on average), which is the closest any of these models comes to a judgement rather than a fixed action; it passes 5/10 of the two rotation levels.

Reasoning mentions (share of episodes): opening 100%, align 100%, width 80%, turn 43%, blocked 32%, sideways 10%, budget 3%

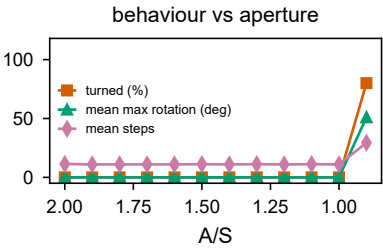
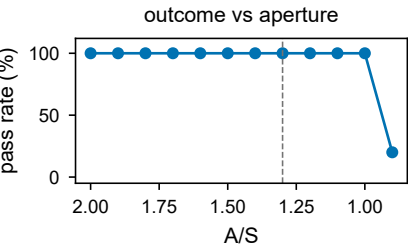
Actions: forward 86.1%, look_down 5.1%, turn_left 3.0%, turn_right 2.0%, right 1.6%, left 1.6%

Strategy labels: FRONTAL 82%, ROT_EARLY 10%, ROT_LATE 8%

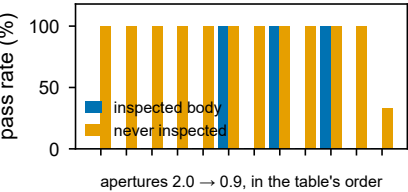
It turned in 18% of all episodes (first turn at x = 3.8 m on average), and in 11% of the episodes at apertures that need no turn at all.

Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 11 apertures positive).

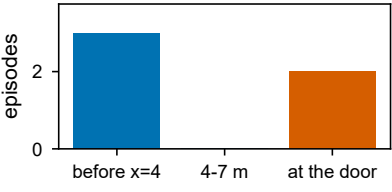
gemini-2.5-flash — straight walker — 60 episodes, overall pass 93%



self-inspection and outcome, within each aperture



when it inspected itself (mean x 4.0 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	11.4	0	0.4	0	—	100
1.9	1.083	5/5	0	0	11.0	0	0.0	0	—	100
1.8	1.026	5/5	0	0	11.0	0	0.0	0	—	100
1.7	0.969	5/5	0	0	11.0	0	0.0	0	—	100
1.6	0.912	5/5	0	0	11.0	0	0.0	0	—	100
1.5	0.855	5/5	0	0	11.2	0	0.2	20	100	100
1.4	0.798	5/5	0	0	11.0	0	0.0	0	—	100
1.3	0.741	5/5	0	0	11.2	0	0.2	20	100	100
1.2	0.684	5/5	0	0	11.0	0	0.0	0	—	100
1.1	0.627	5/5	0	0	11.2	0	0.2	20	100	100
1.0	0.570	5/5	0	0	11.0	0	0.0	0	—	100
0.9	0.513	1/5	80	100	29.4	51	14.4	40	0	33

Type: straight walker. it turns in only 7% of all episodes and passes by walking straight; its score comes from the zero-tolerance frontal pass at A/S 1.0, and at A/S 0.9, where at least 60 degrees is required, it almost always fails: 6/10 over the two rotation levels. Also: never looks at its own body (8%).

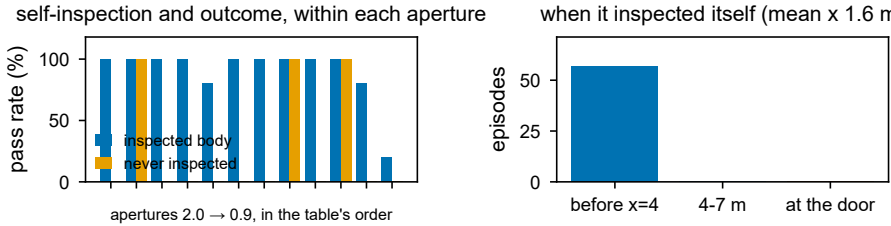
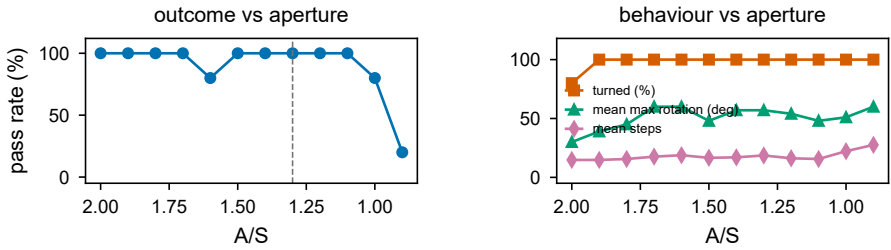
Reasoning mentions (share of episodes): opening 100%, align 92%, turn 32%, blocked 22%, width 12%, sideways 3%

Actions: forward 89.2%, turn_right 2.5%, turn_left 2.2%, look_down 2.1%, left 1.5%, right 1.2%

Strategy labels: FRONTAL 93%, ROT_LATE 5%, ROT_MID 2%

It turned in 7% of all episodes (first turn at x = 7.1 m on average), and in 0% of the episodes at apertures that need no turn at all.

Inspecting its own body, compared inside each aperture: -8.3 percentage points on the pass rate (0 of 4 apertures positive).



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	80	40	14.8	30	3.8	100	100	—
1.9	1.083	5/5	100	40	14.8	39	3.8	80	100	100
1.8	1.026	5/5	100	60	15.6	45	4.6	100	100	—
1.7	0.969	5/5	100	80	17.6	60	6.6	100	100	—
1.6	0.912	4/5	100	75	18.8	60	7.8	100	80	—
1.5	0.855	5/5	100	60	16.6	48	5.6	100	100	—
1.4	0.798	5/5	100	60	17.0	57	6.0	100	100	—
1.3	0.741	5/5	100	40	18.6	57	7.6	80	100	100
1.2	0.684	5/5	100	80	16.2	54	5.2	100	100	—
1.1	0.627	5/5	100	40	15.6	48	4.6	80	100	100
1.0	0.570	4/5	100	50	22.2	51	11.2	100	80	—
0.9	0.513	1/5	100	100	27.6	60	12.6	100	20	—

Type: fixed-policy turner. it turns in 98% of the episodes at apertures that need no turn at all, with a mean maximum rotation of 51 degrees: it is not reading the aperture, it is executing a fixed action, which costs it 6.6 steps over the optimum and still passes only 5/10 of the two rotation levels. Also: inspects its own body constantly (95% of episodes); wastes steps (6.6 over the optimum).

Reasoning mentions (share of episodes): opening 100%, width 100%, turn 98%, sideways 98%, align 95%, blocked 70%, budget 2%

Actions: forward 61.5%, turn_right 20.9%, look_down 8.8%, turn_left 7.1%, right 0.8%, left 0.6%

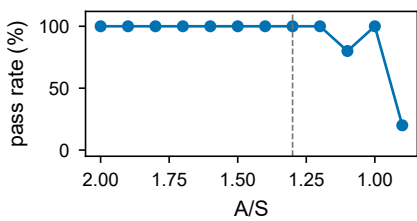
Strategy labels: ROT_EARLY 98%, FRONTAL 2%

It turned in 98% of all episodes (first turn at x = 2.0 m on average), and in 98% of the episodes at apertures that need no turn at all.

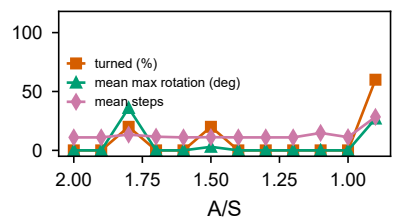
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 3 apertures positive).

glm-4.6v — straight walker — 60 episodes, overall pass 92%

outcome vs aperture



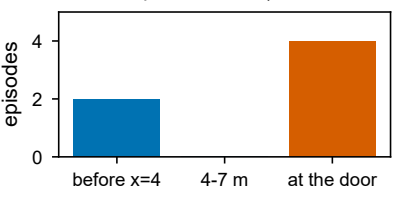
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 5.2 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	11.0	0	0.0	0	—	100
1.9	1.083	5/5	0	0	11.0	0	0.0	0	—	100
1.8	1.026	5/5	20	0	13.4	36	2.4	0	—	100
1.7	0.969	5/5	0	0	11.6	0	0.6	20	100	100
1.6	0.912	5/5	0	0	11.0	0	0.0	0	—	100
1.5	0.855	5/5	20	0	11.2	3	0.2	0	—	100
1.4	0.798	5/5	0	0	11.0	0	0.0	0	—	100
1.3	0.741	5/5	0	0	11.0	0	0.0	0	—	100
1.2	0.684	5/5	0	0	11.0	0	0.0	0	—	100
1.1	0.627	4/5	0	0	14.8	0	3.8	0	—	80
1.0	0.570	5/5	0	0	11.2	0	0.2	20	100	100
0.9	0.513	1/5	60	100	28.4	27	13.4	80	25	0

Type: straight walker. it turns in only 8% of all episodes and passes by walking straight; its score comes from the zero-tolerance frontal pass at A/S 1.0, and at A/S 0.9, where at least 60 degrees is required, it almost always fails: 6/10 over the two rotation levels. Also: never looks at its own body (10%).

Reasoning mentions (share of episodes): align 65%, opening 43%, width 27%, blocked 13%, turn 12%, sideways 3%

Actions: forward 93.7%, turn_right 2.3%, left 1.5%, turn_left 1.1%, look_down 0.8%, right 0.5%

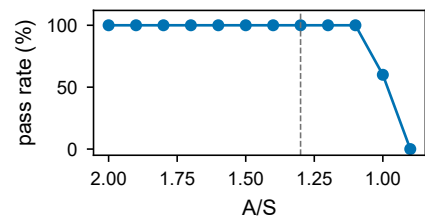
Strategy labels: FRONTAL 90%, ROT_LATE 5%, ROT_EARLY 3%, SIDEWAYS 2%

It turned in 8% of all episodes (first turn at x = 5.0 m on average), and in 4% of the episodes at apertures that need no turn at all.

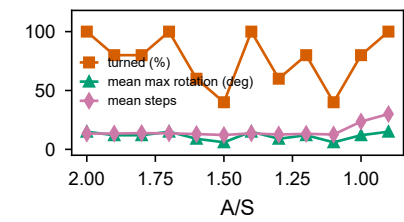
Inspecting its own body, compared inside each aperture: +8.3 percentage points on the pass rate (1 of 3 apertures positive).

gpt-4.1 — fixed-policy turner — 60 episodes, overall pass 88%

outcome vs aperture



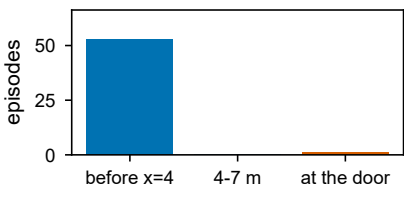
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.4 m)



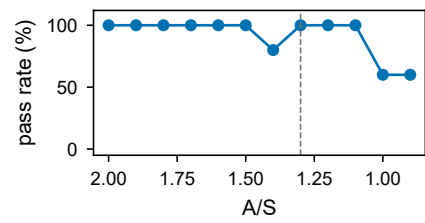
per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	100	0	13.4	15	2.4	100	100	—
1.9	1.083	5/5	80	0	13.4	12	2.4	100	100	—
1.8	1.026	5/5	80	0	13.8	12	2.8	100	100	—
1.7	0.969	5/5	100	0	13.6	15	2.6	100	100	—
1.6	0.912	5/5	60	0	13.0	9	2.0	80	100	100
1.5	0.855	5/5	40	0	12.2	6	1.2	60	100	100
1.4	0.798	5/5	100	0	13.6	15	2.6	100	100	—
1.3	0.741	5/5	60	0	12.6	9	1.6	80	100	100
1.2	0.684	5/5	80	0	13.2	12	2.2	80	100	100
1.1	0.627	5/5	40	0	12.6	6	1.6	80	100	100
1.0	0.570	3/5	80	0	23.6	12	12.6	100	60	—
0.9	0.513	0/5	100	0	30.0	15	15.0	100	0	—

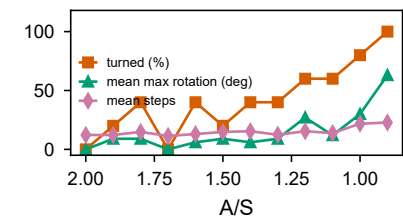
Type: fixed-policy turner. it turns in 75% of the episodes at apertures that need no turn at all, with a mean maximum rotation of 12 degrees: it is not reading the aperture, it is executing a fixed action, which costs it 4.1 steps over the optimum and still passes only 3/10 of the two rotation levels. Also: inspects its own body constantly (90% of episodes). Reasoning mentions (share of episodes): opening 100%, align 98%, width 90%, turn 77%, blocked 23%, sideways 2% Actions: forward 76.2%, look_down 13.2%, turn_left 5.1%, right 1.6%, left 1.3%, look_left 1.3% Strategy labels: ROT_EARLY 70%, FRONTAL 23%, SIDEWAYS 7% It turned in 77% of all episodes (first turn at x = 1.7 m on average), and in 75% of the episodes at apertures that need no turn at all. Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 5 apertures positive).

gpt-4o — turn on demand — 60 episodes, overall pass 92%

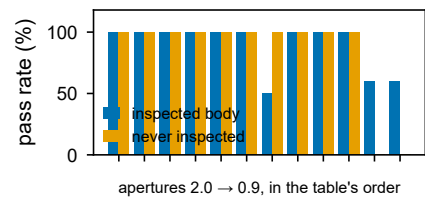
outcome vs aperture



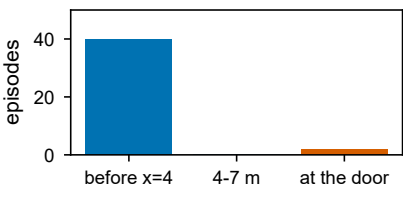
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.6 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	12.2	0	1.2	40	100	100
1.9	1.083	5/5	20	20	12.2	9	1.2	60	100	100
1.8	1.026	5/5	40	0	14.8	9	3.8	80	100	100
1.7	0.969	5/5	0	0	11.4	0	0.4	40	100	100
1.6	0.912	5/5	40	0	13.0	6	2.0	60	100	100
1.5	0.855	5/5	20	20	14.8	9	3.8	80	100	100
1.4	0.798	4/5	40	0	15.4	6	4.4	40	50	100
1.3	0.741	5/5	40	0	12.4	9	1.4	80	100	100
1.2	0.684	5/5	60	20	15.4	27	4.4	80	100	100
1.1	0.627	5/5	60	0	13.8	12	2.8	80	100	100
1.0	0.570	3/5	80	67	21.8	30	10.8	100	60	-
0.9	0.513	3/5	100	100	22.8	63	7.8	100	60	-

Type: turn on demand. whether it turns follows the aperture (only 36% of the no-turn episodes turn, first turn at x = 1.64 m on average), which is the closest any of these models comes to a judgement rather than a fixed action; it passes 6/10 of the two rotation levels.

Reasoning mentions (share of episodes): opening 88%, width 87%, align 85%, turn 45%, blocked 15%

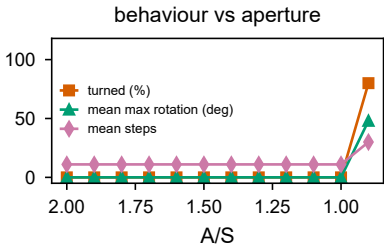
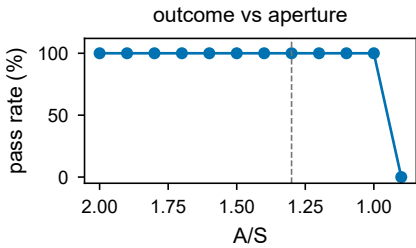
Actions: forward 77.4%, look_down 10.9%, turn_left 6.8%, look_left 1.7%, left 1.0%, look_right 1.0%

Strategy labels: FRONTAL 58%, ROT_EARLY 38%, SIDEWAYS 3%

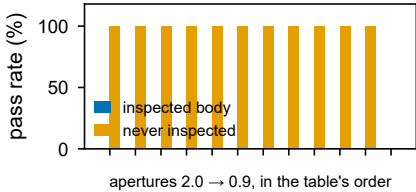
It turned in 42% of all episodes (first turn at x = 1.6 m on average), and in 36% of the episodes at apertures that need no turn at all.

Inspecting its own body, compared inside each aperture: -5.0 percentage points on the pass rate (0 of 10 apertures positive).

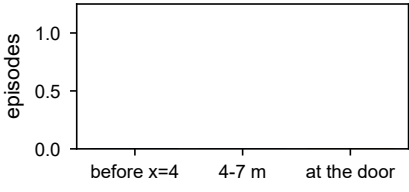
gpt-4o-mini — straight walker — 60 episodes, overall pass 92%



self-inspection and outcome, within each aperture



when it inspected itself (mean x 0.0 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	11.0	0	0.0	0	—	100
1.9	1.083	5/5	0	0	11.0	0	0.0	0	—	100
1.8	1.026	5/5	0	0	11.0	0	0.0	0	—	100
1.7	0.969	5/5	0	0	11.0	0	0.0	0	—	100
1.6	0.912	5/5	0	0	11.0	0	0.0	0	—	100
1.5	0.855	5/5	0	0	11.0	0	0.0	0	—	100
1.4	0.798	5/5	0	0	11.0	0	0.0	0	—	100
1.3	0.741	5/5	0	0	11.0	0	0.0	0	—	100
1.2	0.684	5/5	0	0	11.0	0	0.0	0	—	100
1.1	0.627	5/5	0	0	11.0	0	0.0	0	—	100
1.0	0.570	5/5	0	0	11.0	0	0.0	0	—	100
0.9	0.513	0/5	80	0	30.0	48	15.0	0	—	0

Type: straight walker. it turns in only 7% of all episodes and passes by walking straight; its score comes from the zero-tolerance frontal pass at A/S 1.0, and at A/S 0.9, where at least 60 degrees is required, it almost always fails: 5/10 over the two rotation levels. Also: never looks at its own body (0%).

Reasoning mentions (share of episodes): align 47%, blocked 8%, turn 7%
Actions: forward 89.7%, left 3.6%, right 2.8%, turn_left 2.5%, turn_right 1.3%, look_left 0.1%

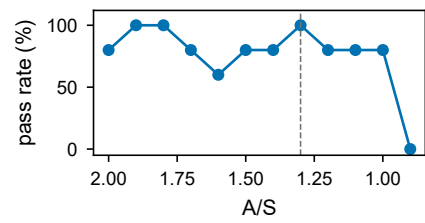
Strategy labels: FRONTAL 92%, ROT_LATE 7%, SIDEWAYS 2%

It turned in 7% of all episodes (first turn at x = 7.2 m on average), and in 0% of the episodes at apertures that need no turn at all.

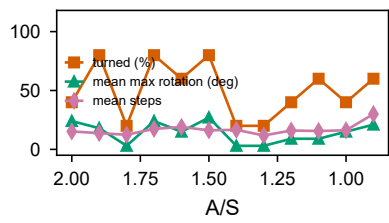
Inspecting its own body, compared inside each aperture: +0.0 percentage points on the pass rate (0 of 0 apertures positive).

qwen-vl-max — mixed — 60 episodes, overall pass 77%

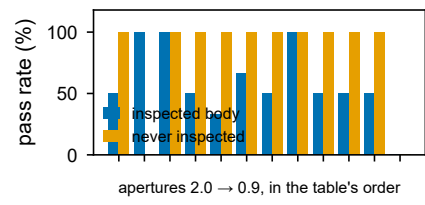
outcome vs aperture



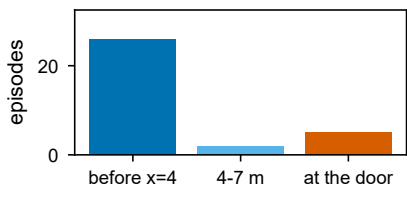
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 2.7 m)

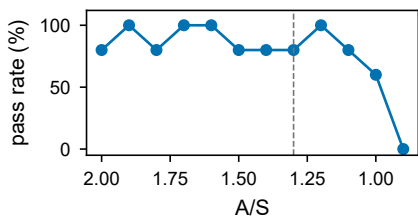


per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

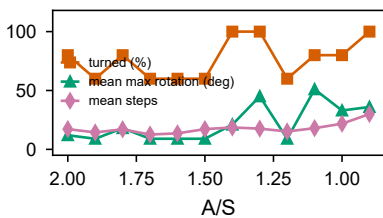
A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	4/5	40	0	15.2	24	4.2	40	50	100
1.9	1.083	5/5	80	0	13.8	18	2.8	100	100	-
1.8	1.026	5/5	20	0	12.4	3	1.4	80	100	100
1.7	0.969	4/5	80	0	17.4	24	6.4	40	50	100
1.6	0.912	3/5	60	0	19.0	15	8.0	60	33	100
1.5	0.855	4/5	80	0	16.2	27	5.2	60	67	100
1.4	0.798	4/5	20	0	16.6	3	5.6	40	50	100
1.3	0.741	5/5	20	0	11.8	3	0.8	20	100	100
1.2	0.684	4/5	40	0	16.0	9	5.0	40	50	100
1.1	0.627	4/5	60	0	15.6	9	4.6	40	50	100
1.0	0.570	4/5	40	25	16.2	15	5.2	40	50	100
0.9	0.513	0/5	60	0	30.0	21	15.0	100	0	-

Type: mixed. the link between turning and aperture is weak: 49% of the no-turn episodes still turn, first turn at x = 2.52 m on average, and it passes 4/10 of the two rotation levels -- between a judgement and a fixed action. Also: wastes steps (5.3 over the optimum); low overall pass rate (77%). Reasoning mentions (share of episodes): align 100%, turn 67%, opening 65%, width 63%, blocked 60%, sideways 20% Actions: forward 72.4%, look_down 8.7%, turn_left 4.5%, right 3.5%, turn_right 3.1%, left 2.6% Strategy labels: FRONTAL 47%, ROT_EARLY 40%, SIDEWAYS 8%, ROT_MID 3%, ROT_LATE 2% It turned in 50% of all episodes (first turn at x = 2.5 m on average), and in 49% of the episodes at apertures that need no turn at all. Inspecting its own body, compared inside each aperture: -40.0 percentage points on the pass rate (0 of 10 apertures positive).

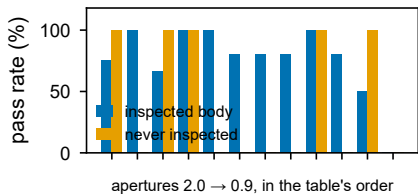
outcome vs aperture



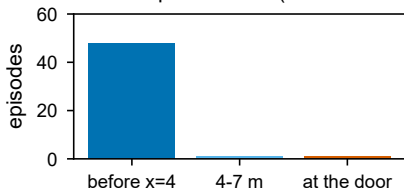
behaviour vs aperture



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.4 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	4/5	80	0	17.2	12	6.2	80	75	100
1.9	1.083	5/5	60	0	14.4	9	3.4	100	100	—
1.8	1.026	4/5	80	0	17.0	18	6.0	60	67	100
1.7	0.969	5/5	60	0	12.6	9	1.6	20	100	100
1.6	0.912	5/5	60	0	13.6	9	2.6	100	100	—
1.5	0.855	4/5	60	0	17.2	9	6.2	100	80	—
1.4	0.798	4/5	100	0	18.6	21	7.6	100	80	—
1.3	0.741	4/5	100	0	17.6	45	6.6	100	80	—
1.2	0.684	5/5	60	0	15.2	9	4.2	80	100	100
1.1	0.627	4/5	80	25	18.0	51	7.0	100	80	—
1.0	0.570	3/5	80	33	21.8	33	10.8	80	50	100
0.9	0.513	0/5	100	0	30.0	36	15.0	80	0	0

Type: fixed-policy turner. it turns in 75% of the episodes at apertures that need no turn at all, with a mean maximum rotation of 22 degrees: it is not reading the aperture, it is executing a fixed action, which costs it 6.4 steps over the optimum and still passes only 3/10 of the two rotation levels. Also: inspects its own body constantly (83% of episodes); wastes steps (6.4 over the optimum); low overall pass rate (78%).

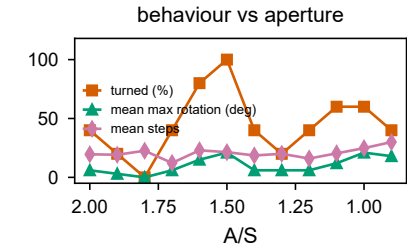
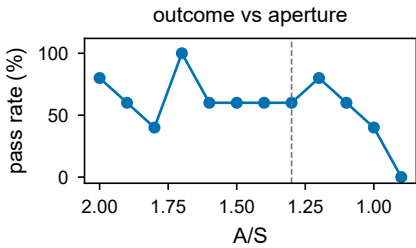
Reasoning mentions (share of episodes): align 100%, opening 97%, turn 97%, width 93%, blocked 85%, sideways 37%, budget 5%

Actions: forward 68.0%, look_down 7.4%, turn_right 7.1%, turn_left 5.6%, look_left 4.1%, backward 2.9%

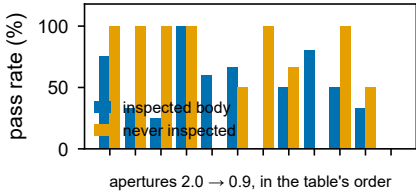
Strategy labels: ROT_EARLY 73%, FRONTAL 23%, ROT_LATE 2%, SIDEWAYS 2%

It turned in 77% of all episodes (first turn at x = 1.8 m on average), and in 75% of the episodes at apertures that need no turn at all.

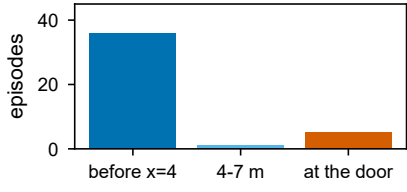
Inspecting its own body, compared inside each aperture: -18.1 percentage points on the pass rate (0 of 6 apertures positive).



self-inspection and outcome, within each aperture



when it inspected itself (mean x 1.7 m)



per aperture (side% = share of that aperture's passes made sideways; '-' = no episode of that kind)

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	4/5	40	0	19.6	6	8.6	80	75	100
1.9	1.083	3/5	20	0	19.2	3	8.2	60	33	100
1.8	1.026	2/5	0	0	22.6	0	11.6	80	25	100
1.7	0.969	5/5	40	0	12.2	6	1.2	40	100	100
1.6	0.912	3/5	80	0	23.0	15	12.0	100	60	—
1.5	0.855	3/5	100	0	21.4	21	10.4	60	67	50
1.4	0.798	3/5	40	0	18.6	6	7.6	40	0	100
1.3	0.741	3/5	20	0	20.0	6	9.0	40	50	67
1.2	0.684	4/5	40	0	16.0	6	5.0	100	80	—
1.1	0.627	3/5	60	0	20.4	12	9.4	80	50	100
1.0	0.570	2/5	60	50	24.8	21	13.8	60	33	50
0.9	0.513	0/5	40	0	30.0	18	15.0	100	0	—

Type: mixed. the link between turning and aperture is weak: 45% of the no-turn episodes still turn, first turn at $x = 1.22$ m on average, and it passes 2/10 of the two rotation levels -- between a judgement and a fixed action. Also: wastes steps (9.3 over the optimum); low overall pass rate (58%). Reasoning mentions (share of episodes): align 100%, opening 87%, width 83%, blocked 80%, turn 68%, sideways 32%, budget 3% Actions: forward 70.3%, look_down 9.4%, right 4.5%, left 3.8%, look_left 3.3%, turn_left 2.7% Strategy labels: FRONTAL 43%, ROT_EARLY 32%, SIDEWAYS 23%, ROT_LATE 2% It turned in 45% of all episodes (first turn at $x = 1.2$ m on average), and in 45% of the episodes at apertures that need no turn at all. Inspecting its own body, compared inside each aperture: -37.0 percentage points on the pass rate (1 of 9 apertures positive).